

SQL-PRACTICE EXAMPLES

2520030021

KARUTURI SAI AKHIL

STEP 1 : CREATE TABLE

The screenshot displays the SQL Developer application window. The top menu bar includes File, Edit, View, Query, Database, Server, Tools, Scripting, and Help. The Navigator pane on the left shows a tree of schemas: bookflow_db, bookflow_week2_db, practice, and sys. The main editor pane shows a script with the following SQL code:

```
1 • CREATE DATABASE SQL_CLG;  
2 • USE SQL_CLG;  
3 • CREATE TABLE orders (  
4     ord_no INT PRIMARY KEY,  
5     purch_amt DECIMAL(10,2),  
6     ord_date DATE,  
7     customer_id INT,  
8     salesman_id INT  
9 );  
10
```

The bottom pane shows the Output window with the following table:

#	Time	Action	Message
✓ 1	11:25:11	CREATE DATABASE SQL_CLG	1 row(s) affected
✗ 2	11:25:24	CREATE TABLE orders (ord_no INT PRIM...	Error Code: 1046. No database selected Selec...
✓ 3	11:25:29	USE SQL_CLG	0 row(s) affected
✓ 4	11:25:33	CREATE TABLE orders (ord_no INT PRIM...	0 row(s) affected

STEP 2: INSERT DATA

Navigator: WEEK1* ddl WEEK2 SQL File 5* x

Limit to 1000 rows

6 ord_date DATE,
7 customer_id INT,
8 salesman_id INT
9);
10
11 • INSERT INTO orders VALUES
12 (70001,150.50,'2012-10-05',3005,5002),
13 (70009,270.65,'2012-09-10',3001,5005),
14 (70002,65.26,'2012-10-05',3002,5001),
15 (70004,110.50,'2012-08-17',3009,5003),
16 (70007,948.50,'2012-09-10',3005,5002),
17 (70005,2400.60,'2012-07-27',3007,5001),
18 (70008,5760.00,'2012-09-10',3002,5001),
19 (70010,1983.43,'2012-10-10',3004,5006),
20 (70003,2480.40,'2012-10-10',3009,5003),
21 (70012,250.45,'2012-06-27',3008,5002),
22 (70011,75.29,'2012-08-17',3003,5007),
23 (70013,3045.60,'2012-04-25',3002,5001);
24

Administration Schemas
Information

No object selected

Output

Action Output

#	Time	Action	Message
✓ 1	11:25:11	CREATE DATABASE SQL_CLG	1 row(s) affected
✗ 2	11:25:24	CREATE TABLE orders (ord_no INT PRIM...	Error Code: 1046. No database selected Select
✓ 3	11:25:29	USE SQL_CLG	0 row(s) affected
✓ 4	11:25:33	CREATE TABLE orders (ord_no INT PRIM...	0 row(s) affected
✓ 5	11:27:26	INSERT INTO orders VALUES (70001,150.50...	12 row(s) affected Records: 12 Duplicates: 0

WHERE CLAUSE

Navigator: WEEK1* ddl WEEK2 SQL File 5* x

Limit to 1000 rows

22 (70011,75.29,'2012-08-17',3003,5007),

23 (70013,3045.60,'2012-04-25',3002,5001);

24

25 • SELECT *

26 FROM orders

27 WHERE purch_amt > 2000;

28

Result Grid

ord_no	purch_amt	ord_date	customer_id	salesman_id
70003	2480.40	2012-10-10	3009	5003
70005	2400.60	2012-07-27	3007	5001
70008	5760.00	2012-09-10	3002	5001
70013	3045.60	2012-04-25	3002	5001
NULL	NULL	NULL	NULL	NULL

orders 1 x

Output

Action Output

#	Time	Action	Message	Duration / Fetch
1	11:25:11	CREATE DATABASE SQL_CLG	1 row(s) affected	0.000 sec
2	11:25:24	CREATE TABLE orders (ord_no INT PRI...	Error Code: 1046. No database selected Sele...	0.000 sec
3	11:25:29	USE SQL_CLG	0 row(s) affected	0.000 sec
4	11:25:33	CREATE TABLE orders (ord_no INT PRI...	0 row(s) affected	0.046 sec
5	11:27:26	INSERT INTO orders VALUES (70001,150.5...	12 row(s) affected Records: 12 Duplicates: 0...	0.016 sec
6	11:28:08	SELECT * FROM orders WHERE purch_amt ...	4 row(s) returned	0.000 sec / 0.0

No object selected

FileEditViewQueryDatabaseServerToolsScriptingHelp

Navigator

WEEK1*ddlWEEK2SQL File 5*

Limit to 1000 rows

SCHEMAS

Filter objects

bookflow_db

bookflow_week2_db

practice

sys

24

25 • SELECT *

26 FROM orders

27 WHERE purch_amt > 2000;

28 • SELECT *

29 FROM orders

30 WHERE ord_date='2012-09-10';

31

Result Grid

Filter Rows:

Edit:

Export/Import:

Wrap Cell Content

	ord_no	purch_amt	ord_date	customer_id	salesman_id
▶	70007	948.50	2012-09-10	3005	5002
	70008	5760.00	2012-09-10	3002	5001
	70009	270.65	2012-09-10	3001	5005
*	NULL	NULL	NULL	NULL	NULL

orders 2 ×

Apply

Output

Action Output

#	Time	Action	Message	Duration / Fetc
✖	2	11:25:24	CREATE TABLE orders (ord_no INT PR...	Error Code: 1046. No database selected Se... 0.000 sec
✔	3	11:25:29	USE SQL_CLG	0 row(s) affected 0.000 sec
✔	4	11:25:33	CREATE TABLE orders (ord_no INT PR...	0 row(s) affected 0.046 sec
✔	5	11:27:26	INSERT INTO orders VALUES (70001,150....	12 row(s) affected Records: 12 Duplicates:... 0.016 sec
✔	6	11:28:08	SELECT * FROM orders WHERE purch_a...	4 row(s) returned 0.000 sec / 0.0
✔	7	11:30:10	SELECT * FROM orders WHERE ord_date...	3 row(s) returned 0.000 sec / 0.0

Administration

SCHEMAS

Information

No object selected

Navigator

SCHEMAS

Filter objects

- bookflow_db
- bookflow_week2_db
- practice
- sys

WEEK1* ddl WEEK2 SQL File 5*

27 WHERE purch_amt > 2000;

28 • SELECT *

29 FROM orders

30 WHERE ord_date='2012-09-10';

31 • SELECT *

32 FROM orders

33 WHERE salesman_id=5001;

Result Grid

ord_no	purch_amt	ord_date	customer_id	salesman_id
70002	65.26	2012-10-05	3002	5001
70005	2400.60	2012-07-27	3007	5001
70008	5760.00	2012-09-10	3002	5001
70013	3045.60	2012-04-25	3002	5001
NULL	NULL	NULL	NULL	NULL

orders 3 x

Output

Action Output

#	Time	Action	Message
3	11:25:29	USE SQL_CLG	0 row(s) affected
4	11:25:33	CREATE TABLE orders (ord_no INT PR...	0 row(s) affected
5	11:27:26	INSERT INTO orders VALUES (70001,150....	12 row(s) affected Records: 12 Duplicate
6	11:28:08	SELECT * FROM orders WHERE purch_a...	4 row(s) returned
7	11:30:10	SELECT * FROM orders WHERE ord_date...	3 row(s) returned
8	11:30:29	SELECT * FROM orders WHERE salesman...	4 row(s) returned

Object Info Session

1 AIRTELPP -1.95%

Search

ORDER BY

WEEK1* ddl WEEK2 SQL File 5* x

Limit to 1000 rows

30 WHERE ord_date='2012-09-10';

31 • SELECT *

32 FROM orders

33 WHERE salesman_id=5001;

34 • SELECT *

35 FROM orders

36 ORDER BY purch_amt DESC;

Result Grid

ord_no	purch_amt	ord_date	customer_id	salesman_id
70008	5760.00	2012-09-10	3002	5001
70013	3045.60	2012-04-25	3002	5001
70003	2480.40	2012-10-10	3009	5003
70005	2400.60	2012-07-27	3007	5001
70010	1983.43	2012-10-10	3004	5006
70007	948.50	2012-09-10	3005	5002
70009	270.65	2012-09-10	3001	5005
70012	250.45	2012-06-27	3008	5002
70001	150.50	2012-10-05	3005	5002
70004	110.50	2012-08-17	3009	5003
70011	75.29	2012-08-17	3003	5007

orders 4 x

Administration Schemas

Information

No object selected

Object Info Session

Output

Action Output

#	Time	Action	Message	Duration / Fetch
4	11:25:33	CREATE TABLE orders (ord_no INT PR...	0 row(s) affected	0.046 sec
5	11:27:26	INSERT INTO orders VALUES (70001,150....	12 row(s) affected Records: 12 Duplicates:...	0.016 sec
6	11:28:08	SELECT * FROM orders WHERE purch_a...	4 row(s) returned	0.000 sec / 0.000 sec
7	11:30:10	SELECT * FROM orders WHERE ord_date...	3 row(s) returned	0.000 sec / 0.000 sec
8	11:30:29	SELECT * FROM orders WHERE salesman...	4 row(s) returned	0.016 sec / 0.000 sec
9	11:31:13	SELECT * FROM orders ORDER BY purch...	12 row(s) returned	0.000 sec / 0.000 sec

File Edit View Query Database Server Tools Scripting Help

Navigator

WEEK1* ddl WEEK2 SQL File 5*

Limit to 1000 rows

SCHEMAS

Filter objects

- bookflow_db
- bookflow_week2_db
- practice
- sys

```

35 FROM orders
36 ORDER BY purch_amt DESC;
37
38
39 • SELECT *
40 FROM orders
41 ORDER BY ord_date;

```

Result Grid

ord_no	purch_amt	ord_date	customer_id	salesman_id
70013	3045.60	2012-04-25	3002	5001
70012	250.45	2012-06-27	3008	5002
70005	2400.60	2012-07-27	3007	5001
70004	110.50	2012-08-17	3009	5003
70011	75.29	2012-08-17	3003	5007
70007	948.50	2012-09-10	3005	5002
70008	5760.00	2012-09-10	3002	5001
70009	270.65	2012-09-10	3001	5005
70001	150.50	2012-10-05	3005	5002
70002	65.26	2012-10-05	3002	5001
70003	2480.40	2012-10-10	3009	5003

orders 5

Administration Schemas

Information

No object selected

Object Info Session

Output

Action Output

#	Time	Action	Message
5	11:27:26	INSERT INTO orders VALUES (70001,150...	12 row(s) affected Records: 12 Duplicates:...
6	11:28:08	SELECT * FROM orders WHERE purch_a...	4 row(s) returned
7	11:30:10	SELECT * FROM orders WHERE ord_date...	3 row(s) returned
8	11:30:29	SELECT * FROM orders WHERE salesman...	4 row(s) returned
9	11:31:13	SELECT * FROM orders ORDER BY purch...	12 row(s) returned
10	11:31:37	SELECT * FROM orders ORDER BY ord_d...	12 row(s) returned

SUM()

File Edit View Query Database Server Tools Scripting Help

Navigator

Filter objects

- bookflow_db
- bookflow_week2_db
- practice
- sys

WEEK1* ddl WEEK2 SQL File 5* x

```

38
39 • SELECT *
40 FROM orders
41 ORDER BY ord_date;
42 • SELECT SUM(purch_amt) AS total_revenue
43 FROM orders;
44

```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: |

	total_revenue
▶	17541.18

Administration Schemas

Information

No object selected

Object Info Session

Result 6 x

Output

Action Output

#	Time	Action	Message	Durat
✓ 6	11:28:08	SELECT * FROM orders WHERE purch_a...	4 row(s) returned	0.000
✓ 7	11:30:10	SELECT * FROM orders WHERE ord_date...	3 row(s) returned	0.000
✓ 8	11:30:29	SELECT * FROM orders WHERE salesman...	4 row(s) returned	0.016
✓ 9	11:31:13	SELECT * FROM orders ORDER BY purch...	12 row(s) returned	0.000
✓ 10	11:31:37	SELECT * FROM orders ORDER BY ord_d...	12 row(s) returned	0.000
✓ 11	11:32:16	SELECT SUM(purch_amt) AS total_revenu...	1 row(s) returned	0.000

AVG

File Edit View Query Database Server Tools Scripting Help

Navigator: WEEK1* ddl WEEK2 SQL File 5* x

SCHEMAS

Filter objects

- bookflow_db
- bookflow_week2_db
- practice
- sys

```

40 FROM orders
41 ORDER BY ord_date;
42 • SELECT SUM(purch_amt) AS total_revenue
43 FROM orders;
44
45 • SELECT AVG(purch_amt) AS average_order
46 FROM orders;

```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: |

average_order
1461.765000

Administration Schemas

Information

No object selected

Object Info Session

Result 7 x

Output

Action Output

#	Time	Action	Message	Dt
✓ 7	11:30:10	SELECT * FROM orders WHERE ord_date...	3 row(s) returned	0.0
✓ 8	11:30:29	SELECT * FROM orders WHERE salesman...	4 row(s) returned	0.0
✓ 9	11:31:13	SELECT * FROM orders ORDER BY purch...	12 row(s) returned	0.0
✓ 10	11:31:37	SELECT * FROM orders ORDER BY ord_d...	12 row(s) returned	0.0
✓ 11	11:32:16	SELECT SUM(purch_amt) AS total_revenu...	1 row(s) returned	0.0
✓ 12	11:32:58	SELECT AVG(purch_amt) AS average_ord...	1 row(s) returned	0.0

MAX()

File Edit View Query Database Server Tools Scripting Help

Navigator: WEEK1* ddl WEEK2 SQL File 5*

SCHEMAS

Filter objects

- bookflow_db
- bookflow_week2_db
- practice
- sys

```

42 • SELECT SUM(purch_amt) AS total_revenue
43 FROM orders;
44
45 • SELECT AVG(purch_amt) AS average_order
46 FROM orders;
47 • SELECT MAX(purch_amt) AS highest_order
48 FROM orders;

```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: |

highest_order
5760.00

Administration Schemas

Information

No object selected

Result 8 x Read

Output

Action Output

#	Time	Action	Message	Duration
8	11:30:29	SELECT * FROM orders WHERE salesman...	4 row(s) returned	0.016 s
9	11:31:13	SELECT * FROM orders ORDER BY purch...	12 row(s) returned	0.000 s
10	11:31:37	SELECT * FROM orders ORDER BY ord_d...	12 row(s) returned	0.000 s
11	11:32:16	SELECT SUM(purch_amt) AS total_revenu...	1 row(s) returned	0.000 s
12	11:32:58	SELECT AVG(purch_amt) AS average_ord...	1 row(s) returned	0.000 s
13	11:33:23	SELECT MAX(purch_amt) AS highest_orde...	1 row(s) returned	0.016 s

MIN()

File Edit View Query Database Server Tools Scripting Help

Navigator: WEEK1* ddl WEEK2 SQL File 5* x

Limit to 1000 rows

SCHEMAS

Filter objects

- bookflow_db
- bookflow_week2_db
- practice
- sys

```

44
45 • SELECT AVG(purch_amt) AS average_order
46 FROM orders;
47 • SELECT MAX(purch_amt) AS highest_order
48 FROM orders;
49 • SELECT MIN(purch_amt) AS lowest_order
50 FROM orders;

```

Result Grid

Filter Rows: Export: Wrap Cell Content: 15

lowest_order
65.26

Administration Schemas

Information

No object selected

Object Info Session

Result 9 x Read Only

Output

Action Output

#	Time	Action	Message	Duration / Fetch
9	11:31:13	SELECT * FROM orders ORDER BY purch...	12 row(s) returned	0.000 sec / 0.000 s
10	11:31:37	SELECT * FROM orders ORDER BY ord_d...	12 row(s) returned	0.000 sec / 0.000 s
11	11:32:16	SELECT SUM(purch_amt) AS total_revenu...	1 row(s) returned	0.000 sec / 0.000 s
12	11:32:58	SELECT AVG(purch_amt) AS average_ord...	1 row(s) returned	0.000 sec / 0.000 s
13	11:33:23	SELECT MAX(purch_amt) AS highest_orde...	1 row(s) returned	0.016 sec / 0.000 s
14	11:33:45	SELECT MIN(purch_amt) AS lowest_order ...	1 row(s) returned	0.000 sec / 0.000 s

GROUP BY

File Edit View Query Database Server Tools Scripting Help

Navigator: WEEK1* ddl WEEK2 SQL File 5* x

Limit to 1000 rows

```

50 FROM orders;
51
52 • SELECT customer_id,
53       SUM(purch_amt) AS total_purchase
54 FROM orders
55 GROUP BY customer_id;
56

```

Result Grid

customer_id	total_purchase
3005	1099.00
3002	8870.86
3009	2590.90
3007	2400.60
3001	270.65
3004	1983.43
3003	75.29
3008	250.45

Administration Schemas

Information

No object selected

Result 10 x Read Only

Output

Action Output

#	Time	Action	Message	Duration / Fetch
✓ 10	11:31:37	SELECT * FROM orders ORDER BY ord_d...	12 row(s) returned	0.000 sec / 0.000 sec
✓ 11	11:32:16	SELECT SUM(purch_amt) AS total_revenu...	1 row(s) returned	0.000 sec / 0.000 sec
✓ 12	11:32:58	SELECT AVG(purch_amt) AS average_ord...	1 row(s) returned	0.000 sec / 0.000 sec
✓ 13	11:33:23	SELECT MAX(purch_amt) AS highest_orde...	1 row(s) returned	0.016 sec / 0.000 sec
✓ 14	11:33:45	SELECT MIN(purch_amt) AS lowest_order ...	1 row(s) returned	0.000 sec / 0.000 sec
✓ 15	11:34:13	SELECT customer_id, SUM(purch_amt)...	8 row(s) returned	0.016 sec / 0.000 sec

GROUP BY + MAX()

File Edit View Query Database Server Tools Scripting Help

Navigator: WEEK1* ddl WEEK2 SQL File 5* x

Limit to 1000 rows

SCHEMAS

Filter objects

- bookflow_db
- bookflow_week2_db
- practice
- sys

```
53 SUM(purch_amt) AS total_purchase
54 FROM orders
55 GROUP BY customer_id;
56 • SELECT customer_id,
57 MAX(purch_amt) AS highest_purchase
58 FROM orders
59 GROUP BY customer_id;
```

Result Grid

	customer_id	highest_purchase
▶	3005	948.50
	3002	5760.00
	3009	2480.40
	3007	2400.60
	3001	270.65
	3004	1983.43
	3003	75.29
	3008	250.45

Administration Schemas

Information

No object selected

Result 11 x

Output

Action Output

#	Time	Action	Message	Duration
✓ 11	11:32:16	SELECT SUM(purch_amt) AS total_revenu...	1 row(s) returned	0.000 sec
✓ 12	11:32:58	SELECT AVG(purch_amt) AS average_ord...	1 row(s) returned	0.000 sec
✓ 13	11:33:23	SELECT MAX(purch_amt) AS highest_orde...	1 row(s) returned	0.016 sec
✓ 14	11:33:45	SELECT MIN(purch_amt) AS lowest_order ...	1 row(s) returned	0.000 sec
✓ 15	11:34:13	SELECT customer_id, SUM(purch_amt...	8 row(s) returned	0.016 sec
✓ 16	11:34:34	SELECT customer_id, MAX(purch_amt...	8 row(s) returned	0.000 sec

Object Info Session

GROUP BY + HAVING

File

Edit

View

Query

Database

Server

Tools

Scripting

Help

SQL

DDL

SQL

SQL

SQL

SQL

SQL

SQL

SQL

SQL

SQL

DDL

SQL

SQL

SQL

SQL

SQL

SQL

SQL

SQL

Navigator

SCHEMAS

Filter objects

bookflow_db

bookflow_week2_db

practice

sys

WEEK1*

ddl

WEEK2

SQL File 5* x

Limit to 1000 rows

Star

Wrench

Search

SQL

59

GROUP BY customer_id;

60

61 •

SELECT salesman_id,

62 SUM(purch_amt) AS total_sales

63 FROM orders

64 GROUP BY salesman_id

65 HAVING SUM(purch_amt) > 3000;

Result Grid

Filter Rows:

Export:

Wrap Cell Content:

	salesman_id	total_sales
▶	5001	11271.46

Administration

Schemas

Information

No object selected

Result 12 x

Output

Action Output

#	Time	Action	Message
✓ 12	11:32:58	SELECT AVG(purch_amt) AS average_ord...	1 row(s) returned
✓ 13	11:33:23	SELECT MAX(purch_amt) AS highest_orde...	1 row(s) returned
✓ 14	11:33:45	SELECT MIN(purch_amt) AS lowest_order ...	1 row(s) returned
✓ 15	11:34:13	SELECT customer_id, SUM(purch_amt...	8 row(s) returned
✓ 16	11:34:34	SELECT customer_id, MAX(purch_amt...	8 row(s) returned
✓ 17	11:35:08	SELECT salesman_id, SUM(purch_amt...	1 row(s) returned

Object Info

Session